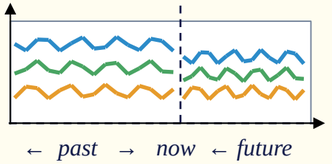
Time-Series Agents: From General to Specific

Scenario & Task Contract

Scenario

- Data streams & context
- Information boundary
- Operational objective



Define requirements

Task Contract

- Output & horizon
- Admissible information
- Constraints & metrics
- Review authority



Select & specialize

Time-Series Agent

LLM-Side Capabilities §3

Temporal Representation & Alignment

Series ↔ language / multimodal representations



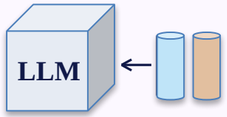
Temporal Understanding & Reasoning

Attributes · Instructions ·Reasoning



Model Adaptation

Training · Adapters · Supervision



Learned temporal capabilities

State-Actions Observations

General Harness (GH) §4

Reusable agent runtime

Interface & Interaction
State · Actions · Observations

Memory & Context
Store · Retrieve · Reuse

Tools & Execution
APIs · Code · Models · Simulators

Control, Verification & Feedback
Check · Retry · Recover

Planning & Workflow
Decompose · Schedule · Stop

Optimization & Evolution
Prompts · Tools ·Memory · Policies

Coordination & Scaling
Roles · Shared state ·Review

General Time-Series Harness (GTS) §5

Cross-dimensional temporal requirements

Temporal Causality
Event time · Availability time
As-of admissibility



Numerical Fidelity
Values · Scales · Units



Time-Indexed State Integrity
Alignment · Missingness · Provenance · Data vintage



Temporal Operation Semantics
Windowing · Frequency · Horizons



Temporal Grounding
Context · Lags · Evidence



Output Auditability
Traces · Checks · Review



Task-Specific Harness (TSK) §6

Task-bound outputs and validation

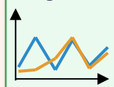
Forecasting & Prediction



Output: Future values / probabilities

Validation: Horizon · Backtesting · Calibration

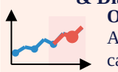
Augmentation & Synthesis



Output: Generated time series

Validation: Temporal fidelity Downstream utility

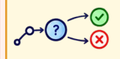
Anomaly Detection & Diagnosis



Output: Anomaly intervals / causes

Validation: Localization False alarms · Evidence

Decision Support



Output: Actions / recommendations

Validation: Utility · Risk Cost · Review

Evaluated by

Layer-Aligned Evaluation §7

Process Validity · GH
Tool-call validity · Recovery
Trace faithfulness
Memory / artifact provenance

Temporal Validity · GTS
As-of / leakage checks · Alignment
Temporal operations · Grounding
Units · Numerical fidelity

Task Performance · TSK
Forecast quality · Synthesis utility
Detection / diagnosis · Decision utility

Resource Behavior
Cost · Latency
Failures · Revisions

Auditable Episode Record
Task · As-of boundary · Visible context
Tool calls · Artifacts · Evidence
Validator decisions · Revisions
Final output · Resource use

Module-Level Evidence Ledger: Claims · Source evidence · Modules · Task contracts · Evaluation · Review status